

MOD 310 Course Outline

MOD 310 — Growth Diagnostics

Course Outline

School	03 Economic Modelling
Level	300 (practitioner / intermediate to advanced)
Hours	27 (indicative band 24–30)
Shape	4 parts · 10 sessions · 6 laboratories · 4 assignments · 1 examination · 1 final project
Delivery	Instructor-led — online, blended or intensive in person
Prerequisites	MOD 302 for the linkage measures and MOD 308 for the multi-regional table these blocks come from. No prior growth theory is assumed, though it helps.
Assessment	Pass 60 %, with 60 % laboratory competency
Version	1.0 · reviewed 2026-08-27 · 12-month cycle

What this course is

Growth diagnostics asks a disciplined question: of all the things wrong with an economy, which one is binding right now? The Hausmann-Rodrik-Velasco decision tree is the standard way to organise the search, and it demands evidence at every branch. This course teaches the framework, then supplies what this school's models can actually contribute to it — a structural profile of eight economies on one common classification — and is exact about the two things it cannot supply: a growth model, and a second point in time.

The course reads every lever the country model exposes — ten families over an intensity ladder — as a growth diagnostic, and the first thing it establishes is which GDP is being read. Deflating reverses the sign of the growth effect for nineteen of the thirty-three solved runs and does so materially for fourteen: a twenty-five per cent import tariff raises GDP at market prices by 18.70 per cent and lowers real GDP by 0.67. No growth statement in this school is made on a nominal aggregate. The course then declines to narrow to a binding constraint, because its own evidence is consistent with three competing explanations and distinguishes none — which under the framework's standard is the finding. All 51 registered findings are in the appendix.

The rule the course is built on

A top-N is a sample. Ask what it was ranked on. Half the diagnostic mistakes in this course's own material came from a table that had already been sorted and filtered before it arrived, and the ranking variable was not the one the reader assumed. The second habit

is the deflator: the same run supports opposite growth conclusions depending on which aggregate is quoted, so both are always shown with the price index between them.

Who it is for

Ministry of Finance and planning-commission economists who must say where growth is constrained, and analysts who consume a growth-diagnostic report and need to know what its evidence actually settles.

Structure

Part	Title	Sessions	Laboratory	Hours
P1	The question, the tree and what counts as evidence	01–03	01	5.7
P2	Structure, and what a profile is not	04–05	02	4.8
P3	Linkages, networks and rankings	06–07	03–04	6.8
P4	Comparison, failure and reporting	08–10	05–06	7.7
	Assignment and assessment time			2.0
	Total	10	6	27.0

The 10 sessions and 6 laboratories, in teaching order, and the subject matter of each:

P1 The question, the tree and what counts as evidence — Sessions 01–03 · 5.7 h · Laboratory 01

- **01 The Growth Diagnostics Question** — Why a list of problems is not a diagnosis; The binding constraint idea; Second-best, and why reform sequencing matters; What the framework replaced; What it asks of the evidence; Where this school’s models can and cannot help
- **02 The Decision Tree** — The top of the tree; Low social returns: geography, human capital, infrastructure; Low appropriability: government and market failures; High cost of finance: domestic and international; Reading a branch as a hypothesis; The branches this school’s data can address
- **03 Evidence at a Branch** — The shadow price test; The movements test; The camels-and-hippos test; The circumvention test; What makes a test decisive rather than suggestive; Why most published diagnostics fail all four
- *Lab 01 What Ran and What Did Not* — What does this extraction contain, and what was it supposed to contain?

P2 Structure, and what a profile is not — Sessions 04–05 · 4.8 h · Laboratory 02

- **04 Structural Change** — Within-sector productivity against reallocation; The McMillan-Rodrik decomposition; Growth-reducing structural change; Premature deindustrialisation; What the decomposition needs, and what we have; Structure as evidence about constraints
- **05 Reading a Structural Profile** — The composition table, column by column; Output share against value-added share; Export share and import penetration; The imported input ratio; Concentration against the even benchmark; What a profile is evidence for
- *Lab 02 A Structural Profile* — What does this economy produce, and how concentrated is it?

P3 Linkages, networks and rankings — Sessions 06–07 · 6.8 h · Laboratory 03–04

- **06 Linkages and Networks** — Backward and forward linkages, restated; The four-way classification; Sectors that sell nothing domestically; Centrality, and what it adds; Thin networks and what they look like; Network evidence at a diagnostic branch
- **07 A Top-N Is Not a Sample** — Three blocks, three ranking variables, none stated; 261 supply drivers, none visible; Maxima in the last row; Selection on a correlate; What a top-N block supports; The check, and its cost
- *Lab 03 Find the Ranking Variable* — What is each top-N block actually a top-N of?
- *Lab 04 Networks and Thin Economies* — What does a thin production network look like?

P4 Comparison, failure and reporting — Sessions 08–10 · 7.7 h · Laboratory 05–06

- **08 Comparing Economies** — The peer set, and why these eight; Concentration across economies; Classification counts across economies; Value added in exports as a position measure; School gap M-10: the empty column; What benchmarking can and cannot establish
- **09 When the Model Does Not Run** — What the SDCGE engine was for; The error, read carefully; What a growth model would have contributed; The counterfactual that does not exist; And the time dimension that does not either; Reporting an analysis with no model in it
- **10 Reporting a Diagnostic** — What a diagnostic note must state; Evidence, inference and recommendation as three sections; How growth diagnostics is misused; The single-constraint fallacy; Reporting a branch you could not test; The diagnostic note template
- *Lab 05 Benchmarking Eight Economies* — What may honestly be compared across these peers?
- *Lab 06 The Diagnostic Note* — What can honestly be said about this economy's constraints?

What a graduate can do

1. State the growth diagnostics question and why it is asked that way.
2. Walk the HRV decision tree and say what evidence each branch needs.
3. Distinguish a binding constraint from a long list of problems.
4. Name the four shadow-price tests and apply them.
5. Read a structural composition table for concentration and position.
6. Read linkage and centrality measures as evidence about production structure.
7. Establish a top-N block's ranking variable before reading any column from it.
8. Say what a single cross-section can and cannot support.
9. Compare eight economies on a common classification without over-reading.
10. State what a failed model run removes from an analysis.
11. Write a growth diagnostic note that separates evidence from inference.

The formal learning outcomes, and the map from each to the assessment that tests it, are in 12_LMS/learning_outcomes.md.

Assessment

Component	Weight	What it is
Knowledge quizzes	—	A bank of 100 items, used for formative practice and drawn on by the theory paper, which carries the marks
Laboratory exercises	34.98 %	6 laboratories, equally weighted. A laboratory is not complete until its validation checks have been read and its named deliverable produced: an irreproducible result is no result
Applied assignments	20 %	What ran, and the framework; Find every ranking variable; Structural evidence at a branch; The diagnostic note
Theory examination	20 %	Closed-book, drawn from the quiz bank and the review questions
Final project	25 %	A complete structural diagnostic for an economy — this one, or one the learner is responsible for — delivered as a seven-section note plus the screening scripts that produced it

The per-laboratory weights are an equal split of the laboratory component rounded to two decimals, so the column sums to 99.98 rather than 100. The component weights are the binding ones.

Pass 60 % overall and 60 % laboratory competency. Merit 70 %. Distinction 80 %. One resubmission of a single failed component within eight weeks. A laboratory is not complete until every validation check has been read and the provenance of every figure recorded: an irreproducible result is no result.

Data and access

No restricted data and no household record. The course reports no growth rate, no projection and no counterfactual: the growth engine returned an error and the structural table is a single year, so neither the shadow-price test nor the movements test can be applied and the course says so at each branch instead of substituting a weaker test. A profile is not a trajectory, and the vocabulary of structural transformation is dynamic where this data is not.

Materials and set-up

A 110-page Technical Manual, 11 lecture decks of 149 slides, 6 laboratory decks of 77 slides, a Laboratory Manual, Instructor Guide and Learner Workbook, a 100-item quiz bank, 5 assessment files, 53

datasets, 4 code exercises, 3 case studies and the LMS package. The README lists every file, and COURSE_BUILD_REPORT.md records how each was produced.

The laboratories run in Python and R against tables that ship with the course; the growth figures are drawn in R and need ggplot2. Rscript is called by full path on this host. No model needs to be instantiated to complete the course, though Laboratory 1 reads the solver log of a run made elsewhere.

The 51 findings registered against this school, and the bearing each has on this course, are carried in the Technical Manual's gap appendix and listed in COURSE_VERSION.json.

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